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Cognitive Reserve Moderates Older Adults' Memory Errors in Autobiographical Reality Monitoring Task

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Abstract

False memory rates differ in individuals with high versus low cognitive reserve and between young-old and old-old age groups. Here we tested how two types of false memory (false alarms to new items and source memory) in two age groups differed with cognitive reserve. Subjects were presented with words and either instructed to generate a past event from their memory associated with the word or to imagine a future event associated with the word. At test, participants were instructed to determine whether the event was a past, future, or new event. Results showed overall false memory rates were lower for young-old adults and those with high reserve. Critically, low cognitive reserve was most associated with source memory errors in young-old but not old-old adults. Reflecting the opposite pattern, false alarms to new items were most associated with low cognitive reserve for old-old, but not young-old adults. These results seem to suggest two different classes of false memories in old age. That is, cognitive reserve was most protective for familiar lures in earlier stages of old age, whereas it was most protective for new lures in later stages of old age. These results support the idea that retrieval monitoring deteriorates with age, potentially due to declines in working memory capacity, but that the decline may be attenuated by cognitive reserve. Furthermore, we suggest that different levels of working memory capacity may be required for monitoring source memory versus item memory, leading to differential effects of cognitive reserve depending upon age.

Keywords

autobiographical memory; false memories; older adults; cognitive reserve; reality monitoring

Introduction

The ability to correctly remember the past is vital to being able to function effectively. However, research has thoroughly established that older adults (typically 60–65+ years of age) possess poorer memory ability than younger adults (typically 18–30 years of age), and that diminished memory in older adults frequently leads to deficits in independent functioning (Cahn-Weiner et al., 2007; Giebel, Challis, & Montaldi, 2015; Overdorp,

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Kessels, Claassen, & Oosterman, 2016). While much aging research focuses on the fact that older adults forget more information than younger adults, memory errors can also take the form of *false memories*. False memories occur when a person believes they experienced an event that they either did not experience or experienced differently from the original event (Roediger & McDermott, 1995). On average, older adults produce about twice as many false memories as younger adults (for a meta-analysis, see McCabe, Roediger, McDaniel, & Balota, 2009). These false memories could create many problems for older adults if, for example, they believe that they paid bills that were not actually paid, took medication that was not actually taken, or made an agreement that they did not actually make.

One explanation for the increase in false memories in older adults is that older adults rely more on familiarity to make decisions about their previous memories than do younger adults (Norman & Schacter 1997; Naveh-Benjamin 2000; Yonelinas, 2002). According to the dual-process model of memory, people decide about the occurrence of previous events by evaluating two types of information—familiarity and recollection. Familiarity refers to an automatic process of memory that provides immediate information about whether the current item has been encountered before, while recollection refers to a more effortful, conscious process that involves the recall of actual details of a previous encounter (Yonelinas, 2002). Behavioral studies have consistently demonstrated that aging selectively impairs recollection-based processes, but spares familiarity-based processes (e.g., Jennings & Jacoby, 1993; Koen & Yonelinas, 2014; Pitarque et al., 2016). Neuroimaging studies have yielded converging evidence with behavioral studies. Older adults exhibit reduced activation of the prefrontal and hippocampal brain regions typically more associated with recollection compared with younger adults (e.g., Cabeza 2002; Logan, Sanders, Morris, & Buckner, 2002; Morcom, Good, Frackowiak, & Rugg, 2003). These reductions in brain activity in recollection-related brain regions are often accompanied by a shift of activation to brain regions associated with familiarity-based processes (e.g., Daselaar, Fleck, Dobbins, Madden, & Cabeza, 2006).

While the differences in memory between younger and older adults have been widely investigated, relatively fewer studies have focused on age differences within the older population. In particular, sometimes distinctions are made between young-old (typically between ages 60–74 years) and old-old (typically above age of 76 years) adults (e.g., McGinnis, 2012; Wang, Kruggel, & Rugg, 2009). Although the specific ages of these two categories differ by study, consistent patterns still appear when comparing young-old and old-old adults. Interestingly, while old-old adults seem to possess consistently poorer source memory than young-old adults, several studies report no difference between these ages for item memory (Cansino, 2009; Cohen & Faulkner, 1989). These findings are in line with the aforementioned declines in recollection and sparing of familiarity. Generally, while both source and item memory may decline with age, old-old adults show greater impairment in source memory compared to item memory (Cansino et al., 2013; Erngrund, Mántylá & Nilsson, 1996). Thus, advanced age seems to be associated with particularly poorer recollection, and thus, the potential for an increased reliance on familiarity compared to young-old adults.

Retrieval Monitoring

While the increased reliance on familiarity over recollection may partially explain the increase in false memories for older adults, the increase may also be due to a decline in retrieval monitoring ability. That is, when attempting to retrieve a particular event, a person may evaluate the contents of the retrieved memory for consistency and plausibility to inform the veracity of the memory signal (Gallo & Lampinen, 2015). Examples of retrieval monitoring include recollection rejection (a.k.a. “recall-to-reject”; e.g. Brainerd, Reyna, Wright, & Mojardin, 2003; Gallo, 2004) and the distinctiveness heuristic (Schacter, Israel, & Racine, 1999; for review, see Gallo, 2013), which involve the rejection of familiar items based on, respectively, recollected items and expectations of a highly distinctive memory trace. Because retrieval monitoring is critical to accurate memory performance (Johnson, Hashtroudi, & Lindsay, 1993), deficits within retrieval monitoring ability may partially explain the increase in false memories experienced by older adults.

The relative impairment of retrieval monitoring in older adults has been shown through both behavioral and neuroimaging studies. For example, when elaboration was used to increase recollected features of experiences, only younger adults were able to take advantage of the increase in recollected features to correspondingly reduce their source memory errors, likely due to older adults possessing deficits in retrieval monitoring (McDonough & Gallo, 2013). Furthermore, while younger adults show greater activation in the DLPFC for more difficult memory decisions (words compared with pictures), which indicates that retrieval monitoring is engaged according to the difficulty of the task, older adults show no difference in DLPFC activation for these different tasks. This evidence may indicate that retrieval monitoring dysfunction is at least partially responsible for the differences in source memory observed between older and younger adults (McDonough, Wong, & Gallo, 2013).

Cognitive Reserve

Although memory dysfunction can be extremely prohibitive to the function of older adults, some recourse can be found for those who have higher cognitive reserve. Cognitive reserve refers to the observed disconnect between the neural pathology associated with aging and dementia and the expected cognitive symptomology. Studies have repeatedly shown that certain characteristics, such as having high socioeconomic status or high levels of education, are related to reduced cognitive symptomology (Stern, 2009). In one well-known study, the linguistic complexity of a person’s writing at a young age was found to predict the cognitive function of that person in old age, as well as their chance of developing Alzheimer’s disease (Snowdon et al., 1996). Many proxies of cognitive reserve have been considered, including educational level, occupational status, and cognitively stimulating leisure activities—all of which possess a significant positive relationship with general cognition in healthy older adults (Opdebeeck, Martyr & Clare, 2016).

However, when specifically assessing how cognitive reserve impacts recollection and familiarity, studies provide mixed evidence. For example, Pitarque and colleagues (2016) discovered that older adults with lower cognitive reserve committed significantly more false alarms and claimed to be “remembered” than those with higher cognitive reserve. This

pattern did not appear for “know” judgments, indicating that cognitive reserve may be important for protecting decline in recollection-based processes with age, but have little or no impact on familiarity-based processes. On the other hand, even in studies investigating familiarity-based strategies, older adults with lower cognitive reserve do sometimes make more false alarms than those with higher cognitive reserve (Sales, Meléndez, Algarabel & Pitarque, 2014). Thus, cognitive reserve may be important for protecting familiarity-based processes as well.

While a rich amount of research has been conducted on false memory within the field of cognitive aging, relatively few studies have specifically investigated the differential false memory rates for tasks assessing item and source memory within older adults by comparing young-old and old-old adults, and to our knowledge, none have investigated how these forms of memory errors might be impacted by cognitive reserve within these age groups. This question is critical to understand how older adults can protect both recollection-based and familiarity-based processes into advanced age.

The Current Study

For the current study, data were reanalyzed from both Experiment 1 and Experiment 2 of McDonough and Gallo (2013) to determine differences between young-old and old-old adults on memory errors within a reality monitoring task, and how cognitive reserve might moderate these errors. While McDonough and Gallo (2013) previously assessed differences in memory accuracy between college-aged younger adults ($M_{\text{age}}=20.37$) and older adults generally ($M_{\text{age}}=76.24$), no analyses comparing memory errors among older adults or cognitive reserve status were previously reported from this data, nor did any previous analyses compare error types between studied vs. non-studied lures. For the current study, we hypothesized that old-old adults would make more memory errors to non-studied lures than young-old adults, as well as more memory errors than young-old adults from misidentifying lures that were originally studied in a different source; however, we also hypothesized that these effects would be ameliorated in older adults with higher cognitive reserve.

Method

Participants

A total of 69 older adults (female $N=49$) participated in the study. A median split was used to separate the participants into young-old ($N=34$; age, 60 – 75) and old-old ($N=35$; age, 76 – 92), an age division similar to that used in previous studies (McGinnis, 2012; Wang et al., 2009). Participants were recruited from the Chicago community and were compensated with \$20 for their time. To ensure suitable mental health, older adults passed a screening for a history of mental illness, showed no signs of depression (Geriatric Depression Scale, young-old $M=3.97$, $SD=3.97$; old-old $M=3.31$, $SD=3.53$) or deficits in cognitive functioning (Mini-Mental State Examination, young-old $M=28.38$, $SD=1.86$; old-old $M=28.43$, $SD=1.38$). Cognitive reserve was measured using the vocabulary portion of the Shipley Institute of Living Scale (SILS; Shipley & Burlingame, 1941), and a median split was used to separate older adults into higher cognitive reserve (SILS >35 , $N=39$) and

lower cognitive reserve ($SILS \leq 35$, $N = 28$). The median split for age and for cognitive reserve were completed independently, resulting in classifications of 18 high reserve young-old individuals, 23 high reserve old-old individuals, 16 low reserve young-old individuals, and 12 low reserve old-old individuals.

Materials

Stimuli used for the current study included 32 positive words from emotional word norms (Bradley & Lang, 1999) and 32 neutral words from McDonough and Gallo (2010). Although there were 64 words in total, 16 words were left out of the study phase for each participant and were instead used as non-studied lures at test. The individual words used for each portion of the study were counterbalanced across participants. All stimuli were presented to participants on a computer through the Cogent2000 toolbox in Matlab.

Procedure

All participants completed the experiment in two phases—a study phase and a test phase that occurred 24 hours later. Before each phase, the experimenter read the instructions aloud to each participant and observed performance to verify a correct understanding of the tasks. During the study phase, each participant was presented with a total of 48 different cue words under two different sets of instructions. For the past event generation block, participants were presented with 32 words and instructed to recall a past memory related to that cue word. For the future event generation block, participants were presented with 32 cue words and were instructed to imagine a future event related to the cue (note that 16 words overlapped between the blocks and were dubbed “both” items, while 16 were unique to each block). For each past or future event, after receiving the cue word, participants were given one minute to recall (past) or imagine (future) as much as they could about the event and verbally provide one-word descriptions of the event, including a title, who was in the event, and the location of the event (these responses were recorded by the experimenter to be used in the test phase).

Although every participant completed both a past event generation block and a future event generation block, the presence of an “elaborative” condition differed according to participant. That is, the study phase was further split into either a past/elaborate event generation or a past/non-elaborate event generation (and the same for future event generation). In the elaborative conditions, participants received an extra 1.5 minutes of elaboration to describe additional sensory details for past events or to describe details of the future plan for the future event. The current analyses collapsed across the elaborative and non-elaborative conditions because we were primarily interested in overall memory errors among different ages and cognitive reserve levels, regardless of the degree of elaboration for individual items (for a detailed discussion of the effects of elaboration, see McDonough and Gallo, 2013).

The test phase occurred 24 hours later. Each participant was presented with a total of 64 events—16 previously generated past events (single-source), 16 previously generated future events (single-source), 16 events generated in both the past and future conditions (“both” items), and 16 new events—before being asked to make a memory judgment. While all

participants had to indicate the source of the events, this question was asked slightly differently depending on participants' retrieval orientation (past or future). Specifically, some participants were asked to indicate whether the event had been previously imagined ("yes" or "no"), while other participants were asked to indicate whether the event was a memory that had been previously recalled ("yes" or "no"). However, our main questions for this paper did not involve the role of retrieval orientation, so we collapsed across the different tests to determine memory accuracy.

On these tests, participants could make two types of memory errors, depending upon the lure with which they were presented: studied or non-studied lures. We define source memory errors as those occurring when participants incorrectly endorsed a studied lure as having been previously generated in the wrong source (e.g., claiming an actual past event was an event generated in the future condition). We define item memory errors as those occurring when participants incorrectly endorsed a non-studied lure as having been previously generated in any source.

Results

Participant Characteristics

A 2 (Age Group: Young-Old, Old-Old) $\times$ 2 (Cognitive Reserve: Higher, Lower) analysis of variance (ANOVA) was conducted on the participant characteristics including sex, education, MMSE, and GDS. Not surprisingly, these ANOVAs resulted in a main effect of Cognitive Reserve group for education, $F(1,69) = 9.56$, $MSE = 6.63$, $p = .003$, $\eta_p^2 = .13$, such that older adults in the Higher Cognitive Reserve group had higher education than older adults in the Lower Cognitive Reserve group. No other main effects or interactions were significant (all p 's $> .09$).

Effects of Targets

Although our main analysis concerned memory errors, we first present the results on hits for completeness. We conducted a 2 (Age Group: Young-Old, Old-Old) $\times$ 2 (Cognitive Reserve: Higher, Lower) $\times$ 2 (Target: Both, Single-source) mixed analysis of variance (ANOVA) on proportion of hits across the memory tests. This analysis resulted in no significant effects (all p 's $> .10$).

Effects of Lures

We next focus on our primary analysis of interest: memory errors. We conducted a 2 (Age Group: Young-Old, Old-Old) $\times$ 2 (Cognitive Reserve: High, Low) $\times$ 2 (Lure: Studied, Non-studied) mixed ANOVA on proportion of memory errors across the criterial recollection tests. This analysis resulted in a main effect of Age Group, $F(1,65) = 4.47$, $MSE = .091$, $p = .038$, $\eta_p^2 = .064$, such that the Young-Old committed fewer memory errors than the Old-Old. A main effect of Cognitive Reserve also was found, $F(1,65) = 12.70$, $MSE = .091$, $p < .001$, $\eta_p^2 = .16$, such that those with higher reserve committed fewer memory errors than those with lower reserve. The analysis also yielded a main effect of Lure, $F(1,65) = 107.63$, $MSE = .039$, $p < .001$, $\eta_p^2 = .62$, such that non-studied lures were associated with fewer memory errors than studied lures. The two-way interactions were not significant (all p 's $> .$

20). Critically, a three-way interaction was found, $F(1,65) = 3.99$, $MSE = .039$, $p = .050$, $\eta_p^2 = .058$.

To better understand the nature of the three-way interaction, we conducted additional analyses between higher and lower cognitive reserve groups, separately for each Age Group and Lure Type (see Figure 1). For studied lures, Young-Old adults with higher cognitive reserve committed fewer memory errors than Young-Old adults with lower cognitive reserve, $t(32) = 3.11$, $MSE = 0.11$, $p = .004$. In contrast, memory errors to studied lures did not differ for Old-Old adults as a function of cognitive reserve, $t(33) = 1.28$, $MSE = 0.11$, $p = .61$. For non-studied lures, no difference was found for memory errors for Young-Old adults as a function of cognitive reserve, $t(32) = 1.91$, $MSE = 0.05$, $p = .065$. In contrast, Old-Old adults with higher cognitive reserve committed fewer memory errors to non-studied lures than Old-Old adults with lower cognitive reserve, $t(33) = 2.35$, $MSE = 0.08$, $p = .025$. Thus, cognitive reserve was most protective against misidentifying studied lures from an alternate source in earlier stages of old age, whereas it was most protective against misidentifying non-studied lures in later stages of old age.

Discussion

The results support our hypothesis that old-old adults commit more memory errors generally than young-old adults. Furthermore, the expectation that older adults with higher cognitive reserve would commit fewer errors than older adults with lower cognitive reserve was also met. Finally, our key finding was that the protective effects of cognitive reserve differed with age such that false memories to studied lures from the incorrect source were reduced in the young-old with higher reserve and false memories to new, non-studied lures were reduced in the old-old with higher reserve.

While previous studies have examined how false memories to studied lures and non-studied lures change with age more generally (e.g. Cansino et al., 2013), our study is the first of which we are aware to examine how cognitive reserve affects the increase in each of these types of false memories according to age among older adults. Our results suggest that cognitive reserve relates to false memory in different ways for people of different age groups. For young-old adults, cognitive reserve maintains prevention of source memory errors, allowing young-old adults to more accurately distinguish between items that were experienced in different contexts (recollection-based processes). In contrast, for old-old adults, cognitive reserve maintains prevention of item memory errors, allowing old-old adults to more accurately identify items that have been not previously experienced (familiarity-based processes).

These findings have many implications for the aging memories of older adults. Although the prefrontal cortex experiences some of the largest changes in structure and function due to aging (Peters, Sethares, & Moss, 1998; Raz et al., 1997; Rossi et al., 2004), cognitive reserve may preserve PFC function. Older adults with low cognitive reserve score lower on tasks that are mediated by the DLPFC such as tasks involving reasoning, working memory, and verbal fluency (Roldán-Tapia, García, Cánovas, & León, 2012). The results of the current study imply that, for older adults with high cognitive reserve, important prefrontal areas such

as the DLPFC may be relatively more intact, which aids in effectively discriminating between targets and lures—hence their lower rate of false alarms. Notably, the benefits of cognitive reserve primarily impacted false alarms, while hits were not affected, a pattern consistent with an effect specific to retrieval monitoring.

Importantly, working memory may play a key role in the process of retrieval monitoring, helping to explain the pattern seen in the current results—namely, the differing effects of cognitive reserve according to age and type of memory process. Considering that source memory is typically a more difficult task than item memory, the shift with age in the type of memory error most affected by cognitive reserve could be due to specific levels of working memory being necessary for each particular memory process. That is, while cognitive reserve seems to protect young-old adults from increasing errors in source memory, after episodic memory declines with age, old-old adults may no longer possess the working memory capacity for accurate source memory regardless of cognitive reserve status. This idea is consistent with findings that source memory decreases more rapidly with age than item memory (Erngrund, Mántylä & Nilsson, 1996). In contrast, young-old adults may retain sufficient levels of working memory capacity necessary for item memory regardless of cognitive reserve status, but after cognitive decline with advanced age, the working memory capacity of old-old adults necessary for item memory may be impaired unless this process is attenuated due to cognitive reserve.

This line of reasoning is supported by our observation that old-old adults with high cognitive reserve somewhat resemble young-old adults with low cognitive reserve for both item and source memory errors (see Figure 1), indicating the potential that cognitive reserve leads to a general attenuation of memory decline with age.

The present results could also be considered as fitting within the activation-monitoring framework for false memories (Roediger, Watson, McDermott, & Gallo, 2001; McDermott & Watson, 2001). When considering the roles of recollection and familiarity within the activation/monitoring framework, separate memory traces may be monitored differently, resulting in different neural requirements. For example, more difficult, previously studied lures have led to higher levels of PFC activation than relatively simpler, non-studied lures (McDonough et al., 2013). Evidence from neuroimaging studies are consistent with the idea that monitoring of recollection-based memory traces is more difficult than monitoring of familiarity-based memory traces, and may depend more heavily upon working memory capacity (Oberauer, 2005). Thus, the consequences of cognitive reserve's relative preservation of working memory capacity may differ according to the neural resources needed for a particular type of retrieval monitoring.

Finally, the complex interaction that we found between age, cognitive reserve, and type of memory error may partially explain findings within the literature regarding cognitive reserve that previously appeared contradictory. Pitarque and colleagues (2016) showed that older adults of higher and lower cognitive reserve status possessed relatively equal false alarms for “know” judgments, indicating that familiarity-based processes may not be protected by cognitive reserve. However, Sales and colleagues (2014) showed that those with low cognitive reserve made more false alarms than those with high cognitive reserve. In light of

the interaction from the current study, the discrepancy between these two studies may be at least partially explained by the fact that older adults in the study by Sales and colleagues were, on average, older than the older adults within the study by Pitarque and colleagues. Thus, level of cognitive reserve may have been more relevant to the older sample within the study by Sales and colleagues, contributing toward the effect that was shown with respect to familiarity-based processes.

Conclusion

The current study underscores the importance of cognitive reserve for minimizing false memories later in life. Item and source memory are foundational cognitive abilities that make possible the ability to appropriately care for oneself and to meaningfully relate to the experiences of others, leading to a more fulfilling life. Thus, as much as people are able to build cognitive reserve in preparation for advanced age, they should do so in order to maintain optimal quality of life. Furthermore, the current study suggests that cognitive reserve may preserve retrieval monitoring abilities in older adults. As the aging population increases, the ability to predict the difficulties that specific older adults are likely to encounter is increasingly important to provide appropriate care for those individuals. Thus, knowledge of the specific cognitive abilities likely to be affected may be of use to physicians, families, and care providers for the advanced aging population. For example, adults over 75 with low cognitive reserve appear to be at the greatest risk for losing the ability to distinguish between events that did or did not occur. The current examination of the effects of cognitive reserve may therefore contribute not only to the understanding of how memory changes with age, but also to the ability to live well into advanced age.

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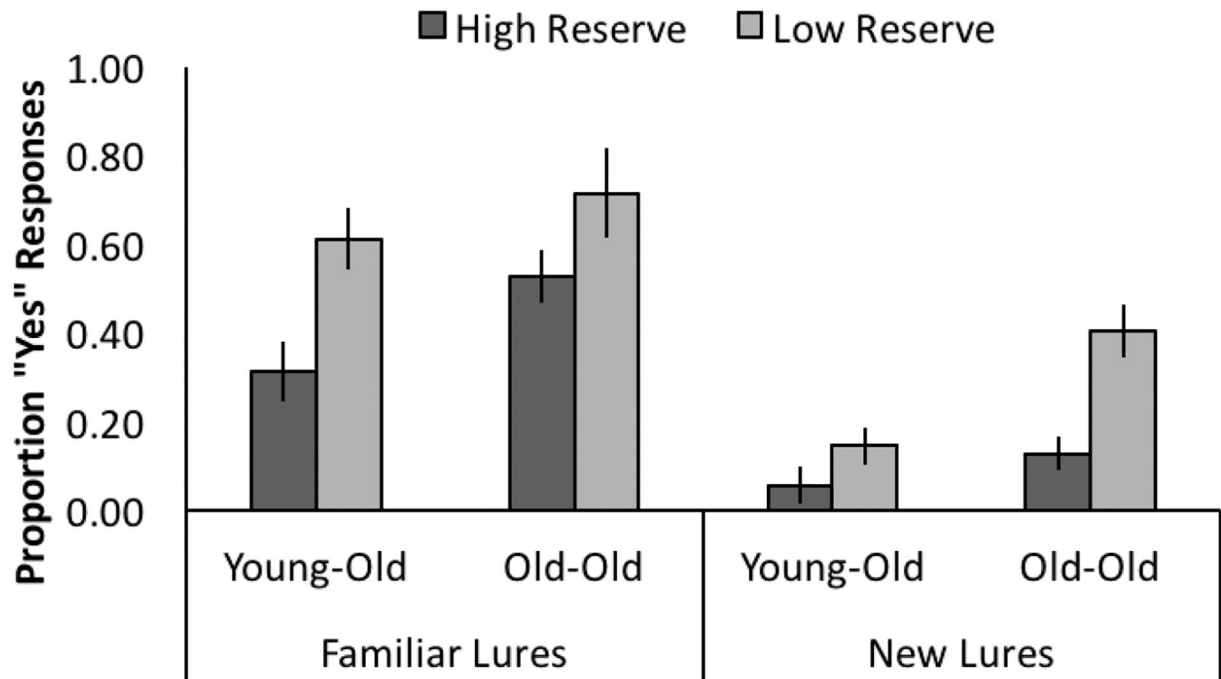


Figure 1.

The proportion of “yes” responses across criterial recollection tests for familiar lures (left) and new lures (right). Older adults with high cognitive reserve are shown in dark gray bars and older adults with low cognitive reserve are shown in light gray bars. Young-Old adults with high cognitive reserve committed fewer memory errors to familiar lures than Young-Old adults with low cognitive reserve, whereas memory errors did not differ for Old-Old adults as a function of cognitive reserve. For new lures, no difference was found for memory errors to new lures for Young-Old adults as a function of cognitive reserve, whereas Old-Old adults with high cognitive reserve committed fewer memory errors to new lures than Old-Old adults with low cognitive reserve. Error bars represent standard error of the mean.